Predictive Flight Fare Dynamics: Identifying Optimal Booking Horizons and Date-Specific Pricing

Group name	Prompters (Text / Documents, Tabular / Transactional, Finance / Economics)
Dataset	kaggle.com/datasets/shubhambathwal/flight-price-prediction
Code	github.com/Viren286/DSC_PROJECT (notebook `main_v2.ipynb`, `hypothesis_tests.py` and README)

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4	Madhav Raj Gaur	2024323	madhav24323@iiitd.ac.in	Tests 1–3 (5)
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1. Dataset

Source: Kaggle, "Flight Price Prediction": flight listings scraped from the EaseMyTrip booking portal for the six largest Indian metros (Delhi, Mumbai, Bangalore, Kolkata, Hyderabad, Chennai) over a 50-day window. One row is one flight listing on one search date.

What is raw about it: a redundant row index (`Unnamed: 0`); ordinal and nominal attributes stored as text (`departure_time`, `stops`, `class`); a right-skewed, bimodal target (Rs. 1,105 to Rs. 123,071); and a flight-code string with 1,561 levels.

	Raw file	After encoding
Instances (rows)	300,153	297,920
Attributes (columns)	12	1,629
Instances × attributes	3,601,836	485,311,680

Size check: $297{,}920 \times 1{,}629 = 485{,}311{,}680 \geq 3{,}000{,}000$, so the requirement is met (the raw file alone has $300{,}153 \times 12 = 3{,}601{,}836$). One-hot encoding of the 1,561 flight codes and 30 engineered routes is what takes it to 1,629 columns. The modelling matrix after dropping `flight` is $297{,}920 \times 68 = 20{,}258{,}560$, still above 3M.

Key features

Feature	Type	Real-world meaning
`price`	continuous	ticket fare in INR; our response
`days_left`	discrete	days between booking and departure (1–49)
`duration`	continuous	total travel time in hours, including layovers
`class`	ordinal	Economy < Business seat
`airline`	nominal	operating carrier, 6 levels
`departure_time`	ordinal	time-of-day slot, 6 levels
`stops`	ordinal	zero < one < two or more

2. Data preparation

Issue found	What we did	Why, and not the alternative
Redundant index `Unnamed: 0`	dropped	a running row number carries no information and would leak row order into a model
Missing values, duplicates	checked: 0 missing, 0 duplicate rows	nothing to impute or drop
2,110 `duration` values above the IQR fence (30.18 h)	rows dropped ($Q_3 + 1.5\,\mathrm{IQR}$ rule)	long-layover itineraries inflate fare variances, which Test 3 compares. Capping would invent durations that were never flown
123 `price` values above the fence (Rs. 99,128)	rows dropped	0.04% of rows, all extreme Business fares, that would dominate the variance tests; a log transform would change the scale of every test
`flight` has 1,561 levels	dropped before modelling (Fig. 1j)	1,561 sparse dummy columns, and the code prefix is fully determined by `airline`
Text categoricals	one-hot encoded; added `route` = source→destination	linear models need numbers; label codes would impose a false order on airlines and cities

3. What changed after preparation

Measure	Before	After
Rows	300,153	297,920 (2,233 dropped, 0.74%)
Attributes	12	11 raw; 1,629 encoded; 68 for modelling
Missing / duplicate rows	0 / 0	0 / 0
Duration range, mean	0.83–49.83 h, 12.22 h	0.83–30.17 h, 12.07 h
Price range, median	Rs. 1,105–123,071, Rs. 7,425	Rs. 1,105–98,972, Rs. 7,424
Economy share	68.85%	68.74%

Dropping 0.74% of rows cut the longest trip from 49.8 h to 30.2 h and the highest fare from Rs. 123k to Rs. 99k, while the median fare moved by Rs. 1 and the class mix by 0.1 percentage points. Dropping `flight` shrank the modelling matrix from 1,629 to 68 columns without losing airline information.

4. Basic questions and problem statement

Basic questions

1. Which airline is cheapest? AirAsia (Rs. 4,091 on average); Vistara is dearest (Rs. 30,369).
2. How far apart are Economy and Business? Rs. 6,530 vs. Rs. 52,446 on average, an $8\times$ gap that makes the fare distribution bimodal.
3. Does a longer trip cost more? Weakly: Pearson $r = 0.215$ (and $r = -0.091$ for `days_left`).
4. Which departure slot is dearest? Night (Rs. 23,037); Late Night is cheapest (Rs. 9,295).
5. Which routes are busiest? Delhi→Mumbai and Mumbai→Delhi, with median fares of Rs. 7,262 and Rs. 6,300.

Our problem statements

P1 (primary, regression). How does the fare of a flight change with the number of days left before departure, and after how many days does booking earlier stop saving money (the booking "sweet spot")? Does the last day bring a drop?

P2. Are late bookings (≤ 20 days) more expensive *and* more volatile than early bookings (> 20 days), and is that explained by a different class mix?

How they are novel: the basic questions describe static averages of one attribute at a time. Ours follow the fare *trajectory* across the booking horizon, locate where the advance-purchase saving stops, and separate the horizon effect from the class mix, which drives most of the raw variance.

5. Statistical inferences

Rows tested: all 297,920 cleaned listings. Late $=$ `days_left` ≤ 20 (112,403 rows), early > 20 (185,517); last-minute $=\leq 7$ days. Every plan in part a) was fixed before the test was run. With five tests we also check each decision at the Bonferroni level $0.05/5 = 0.01$; none changes. With $N \approx 3\times10^5$ almost any gap is significant, so we judge each test by whether its 95% CI clears the smallest difference that matters.

#	Question	Statistic	Decision	Matters?
1	Do fares drop on the last day?	$t = 14.96$	reject H_0	yes
2	Are late bookings dearer?	$Z = 46.09$	reject H_0	yes
3	Are late fares more volatile?	$F = 0.977$	fail to reject	no (pooled)
4	Does fare fall with days left?	$t = -49.77$	reject H_0	yes
5	Is class mix tied to timing?	$\chi^2 = 46.89$	reject H_0	no

Test 1 (P1): do fares drop on the last day?

a) Hypotheses, written before running the test

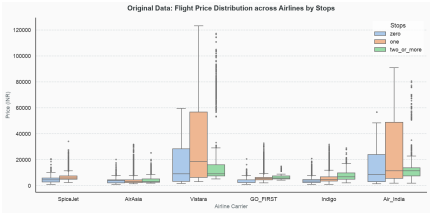
$\boldsymbol{H_0}$	$\mu_2 - \mu_1 \leq 0$
$\boldsymbol{H_1}$	$\mu_2 - \mu_1 > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	Welch t-test
Smallest difference that matters	Rs. 1,000

μ_2, μ_1: mean fare with 2 days and 1 day left. One-sided, because Fig. 1g suggests a last-day clearance drop. Rs. 1,000 is about 5% of a typical ticket, a saving a traveller would notice.

b) Other tests we considered

Two independent groups (3,942 and 1,884 listings), a continuous response and unknown σ call for a two-sample t-test. We use Welch's version because the spreads differ (SD Rs. 24,138 vs. Rs. 18,841; Levene $p = 10^{-50}$). With $n > 1{,}800$ per group the CLT makes the mean difference approximately normal despite the skew.

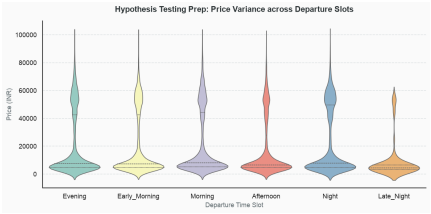
Also considered	Why we did not use it
Pooled t-test	assumes equal variances, which Levene rejects. Run anyway: $t = 13.72$ (5,824), $p = 2\times10^{-42}$, same decision
Mann–Whitney U	tests a shift in the whole distribution, not the mean fare paid. Agrees: $p = 4\times10^{-24}$, $P(\text{day 2} > \text{day 1}) = 0.58$, medians Rs. 17,854 vs. Rs. 15,078
Permutation test	assumption-free check of the mean gap; 10,000 shuffles give $p < 10^{-4}$
Bootstrap CI	assumption-free CI for the gap: [Rs. 7,542, Rs. 9,759], matching Welch



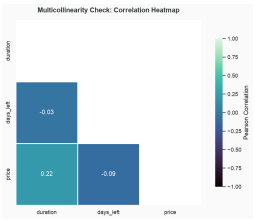
(a) Vistara and Air India one-stop fares have the widest, highest spread; budget carriers stay below Rs. 35k.



(b) Fares form two bands (Economy, Business) and the fitted line slopes gently down with days left.



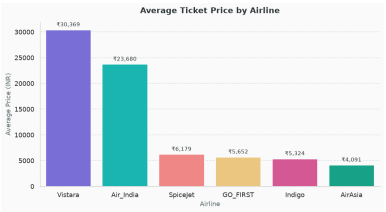
(c) Every departure slot is bimodal; Late Night lacks most of the Business mode.



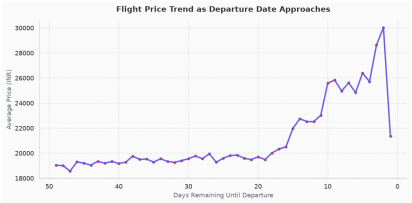
(d) Numeric features are only weakly correlated: no multi-collinearity.



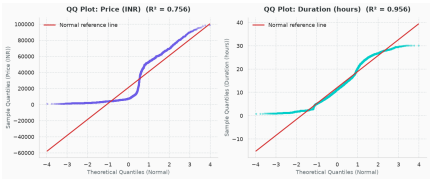
(e) 68.7% of listings are Economy, 31.3% Business.



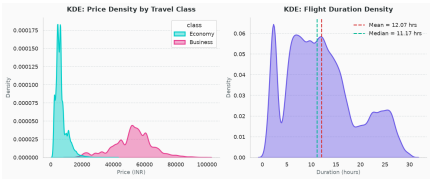
(f) Vistara and Air India average 4–7× the fare of low-cost carriers.



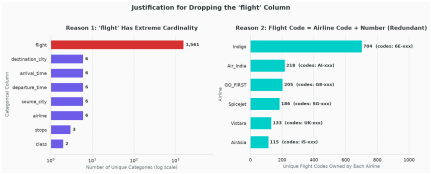
(g) Mean fare is flat to about 20 days out, climbs steeply, peaks 2 days out, then drops on the last day.



(h) Price is far from normal (QQ $R^2 = 0.756$), so every parametric test gets a rank-based or resampling check.



(i) Price density splits cleanly by class; duration is multimodal (mean 12.07 h).



(j) `flight` has 1,561 levels and its code prefix maps one-to-one to `airline`, so we drop it.

Figure 1: Exploratory data analysis of the cleaned data ($N = 297{,}920$). Each panel has its one-line takeaway.

c) Result

Estimate	Rs. 8,672 (Rs. 30,024 vs. Rs. 21,351)
95% confidence interval	Rs. 7,535 to Rs. 9,809
Test statistic (d.f.)	$t = 14.96$ (4,634)
p-value	1.0×10^{-49}
Decision	reject H_0

Fares drop by about Rs. 8.7k on the final day (Cohen's $d = 0.38$). The whole interval is more than seven times the Rs. 1,000 threshold, so the drop matters (Fig. 2a).

Test 2 (P2): are late bookings more expensive?

a) Hypotheses, written before running the test

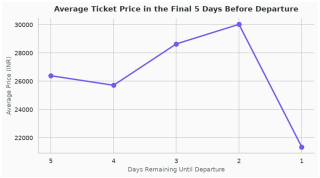
H_0	$\mu_{\text{late}} - \mu_{\text{early}} = 0$
H_1	$\mu_{\text{late}} - \mu_{\text{early}} \neq 0$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	two-sample z-test
Smallest difference that matters	Rs. 500

Two-sided, because a late *discount* (as on the last day) was also plausible.

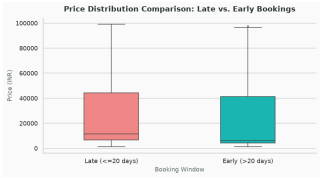
b) Other tests we considered

Both groups are very large (112,403 and 185,517), so the sample SDs estimate σ almost exactly and the z statistic is valid without assuming normality.

Also considered	Why we did not use it
Welch t-test	identical at this size: $t = 46.09$ (239,268); adds nothing
Mann–Whitney U	checks the result is not driven by outliers: $p < 10^{-300}$, $P(\text{late} > \text{early}) = 0.65$, medians Rs. 11,511 vs. Rs. 6,067
Paired t-test	nothing to pair: listings are not matched across windows
One-way ANOVA over 49 days	answers a vaguer question ("some day differs"); Test 4 models the trend instead
z-test within each class	run as a check, since class dominates price (result below)



(a) Test 1: mean fare peaks 2 days out and falls sharply on the final day.



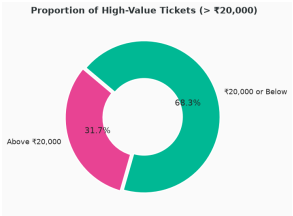
(b) Test 2: late bookings have a higher median and upper quartile.



(c) Test 4: the fitted line falls by about Rs. 152 per extra day booked ahead.



(d) Late Night fares sit lowest; Night and Morning highest.



(e) 31.7% of tickets cost over Rs. 20,000, nearly all Business.



(f) Test 3: pooled fare SD is almost the same for early and late bookings.



(g) Test 5: the Business share is only slightly higher for last-minute bookings.

Figure 2: Plots for the hypothesis tests. Each panel has its one-line takeaway.

c) Result

Estimate	Rs. 3,924 (Rs. 23,327 vs. Rs. 19,402), +20.2%
95% confidence interval	Rs. 3,757 to Rs. 4,091
Test statistic (d.f.)	$z = 46.09$ (none for a z-test)
p-value	$< 10^{-300}$
Decision	reject H_0

Late bookings cost about Rs. 3.9k more, far above Rs. 500 (Fig. 2b). The premium is Rs. 4,069 [4,036, 4,102] (+81%) within Economy and Rs. 1,846 [1,673, 2,018] (+3.6%) within Business. Narrowing "late" to the last week widens the gap to Rs. 6,912 ($z = 49.4$).

Test 3 (P2): are late-booking fares more volatile?

a) Hypotheses, written before running the test

H_0	$\sigma^2_{\text{late}}/\sigma^2_{\text{early}} \leq 1$
H_1	$\sigma^2_{\text{late}}/\sigma^2_{\text{early}} > 1$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	F-test for two variances
Smallest difference that matters	SD 10% higher (variance ratio 1.21)

b) Other tests we considered

P2 asks whether the late variance is *larger*; the F ratio is exactly that, and it allows a one-sided alternative and a CI for the ratio. It assumes normality, which fails (Fig. 1h), so we compared it with robust tests.

Also considered	Why we did not use it
Bartlett	also assumes normality: $\chi^2 = 18.6$, $p = 2 \times 10^{-5}$
Levene (mean)	robust but two-sided only: $W = 134.2$ (1, 297,918), $p = 5 \times 10^{-31}$
Brown–Forsythe (median)	most robust, two-sided only: $W = 169.9$, $p = 8 \times 10^{-39}$; mean $\lvert x - \text{median}\rvert$ is Rs. 16,328 (late) vs. Rs. 15,309 (early)
F-test within each class	run as a check, since the class gap dominates the pooled spread

c) Result

Estimate	variance ratio 0.977 (SD Rs. 22,430 vs. Rs. 22,690)
95% confidence interval	0.967 to 0.988
Test statistic (d.f.)	$F = 0.977$ (112,402; 185,516)
p-value	1.000
Decision	fail to reject H_0

Pooled over both classes, late fares are not more variable: the interval lies below 1, far from 1.21 (Fig. 2f). The tests disagree because they measure different things, and pooling hides real volatility. Within Economy, late fares have more than twice the SD (Rs. 4,425 vs. Rs. 2,030; $F = 4.75$ [4.69, 4.81]); within Business they are about 10% more spread ($F = 1.21$ [1.19, 1.23]); both $p < 10^{-90}$. P2's answer is *yes within each class, no when pooled*.

Test 4 (P1): does fare fall as booking moves earlier?

a) Hypotheses, written before running the test

H_0	$\beta_1 = 0$ in price $= \beta_0 + \beta_1$ days_left $+ \varepsilon$
H_1	$\beta_1 \neq 0$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	t-test on the OLS slope
Smallest difference that matters	Rs. 100 per day (Rs. 1,000 for 10 days)

b) Other tests we considered

P1 is a regression problem; the slope is the average saving per extra day of advance booking, and its t-test is the standard inference for it.

Also considered	Why we did not use it
HC1 robust SE	residuals are heteroscedastic; robust SE 3.06 vs. 3.05, $t = -49.7$, same decision
Spearman, Kendall	test any monotone trend, not the size of the saving: $\rho = -0.266$, $\tau_b = -0.185$, both $p < 10^{-300}$; much stronger than Pearson $r = -0.091$, so the trend is monotone but not linear
Regression with class	checks the slope is not a class artefact: $\hat{\beta}_1 = -130.6$ (SE 1.03), $R^2 = 0.887$
Piecewise slopes	test the sweet spot directly (result below)

c) Result

Estimate	−Rs. 152 per extra day booked ahead
95% confidence interval	-157.9 to -146.0
Test statistic (d.f.)	$t = -49.77$ (297,918)
p-value	$< 10^{-300}$
Decision	reject H_0

The whole interval is beyond Rs. 100/day (Fig. 2c). Pooled R^2 is only 0.008 because class dominates; within Economy it is 0.313. The piecewise fit locates the sweet spot: within 20 days of departure each extra day booked ahead saves Rs. 464 [440, 488], but beyond 20 days only Rs. 25 [13, 38], under the Rs. 100 threshold. **Booking about three weeks ahead captures nearly all the savings.**

Test 5 (P2): does the class mix change for last-minute bookings?

a) Hypotheses, written before running the test

H_0	booking timing (last-minute ≤ 7 days / standard) is independent of class
H_1	the two are associated
Significance level	$\alpha = 0.05$
Side	right tail of χ^2
Test	Pearson χ^2 test of independence
Smallest difference that matters	5 percentage points in Business share

This checks whether the premium in Tests 1–2 could simply be more Business tickets among late bookings.

	Business	Economy	Business share
Last-minute (≤ 7 d)	10,422	21,211	32.95%
Standard (> 7 d)	82,706	183,581	31.06%

b) Other tests we considered

Two categorical variables and all expected counts $\geq 9{,}888$, so the χ^2 approximation is excellent.

Also considered	Why we did not use it
Yates-corrected χ^2	correction matters only for small counts: $\chi^2 = 46.80$, same decision
G-test	likelihood-ratio version, equivalent at this size: $G = 46.50$, $p = 9 \times 10^{-12}$
Two-proportion z-test	equivalent ($z = 6.85$, $z^2 = \chi^2$); we use it for the CI below
Fisher's exact test	built for small expected counts, so unnecessary: OR $= 1.09$, $p = 9 \times 10^{-12}$

c) Result

Estimate	+1.89 pp Business share (Cramér's $V = 0.013$)
95% confidence interval	+1.34 to +2.43 pp
Test statistic (d.f.)	$\chi^2 = 46.89$ (1)
p-value	7.5×10^{-12}
Decision	reject H_0

The association is real but does not matter: the whole interval is below 5 pp (Fig. 2g). So the late-booking premium in Tests 1–2 is not a class-mix effect.

What we conclude

Fares are flat beyond about 20 days and rise steeply inside that window, so about three weeks ahead is the sweet spot (Test 4). Late bookings cost Rs. 3.9k more, +81% in Economy (Test 2), yet fares drop by Rs. 8.7k on the very last day (Test 1). Late fares are more volatile within each class, which a pooled test misses (Test 3), and none of this is a class-mix effect (Test 5).

6. Limitations

- **Association, not cause.** Fare gaps reflect demand, competition and seat inventory, none of which is recorded; the tests cannot show that booking later *causes* higher fares.
- **Listings are not fully independent.** The same flight appears on many search dates, so the effective sample is smaller than 297,920 and our p-values are optimistic. The effect sizes and intervals are less affected.
- **One 50-day window.** Seasonality, holidays, fuel surcharges and load factors are absent, so the 20-day sweet spot may differ in other seasons.
- **Dropped rows.** The 2,233 outlier rows (long layovers and extreme Business fares) are not covered by our conclusions.
- **Bimodal fares.** Business fares dominate the pooled variance and mask within-class effects (Tests 3–4); the P1 model must condition on class or use $\log(\text{price})$.